

Welcome to Week 3: Advice for Applying Machine Learning

You now have a toolbox of powerful algorithms (linear/logistic regression, neural networks). This week's goal is to learn how to **use these tools effectively** and efficiently.

The key to being an effective machine learning practitioner is making good decisions about **what to do next** to improve your model's performance.

The Core Problem: What to Try Next? 🤔

Imagine you've built a model (e.g., regularized linear regression to predict housing prices), but it makes **unacceptably large errors**.

You have many options. What should you try?

- Get more training examples.
 - Try a smaller set of features.
 - Try getting *more* (additional) features.
 - Try adding polynomial features (e.g., x_1^2 , x_1x_2).
 - Try decreasing the regularization parameter, λ .
 - Try increasing the regularization parameter, λ .
-

The Challenge: Wasting Time ⌚

- Some of these options are extremely time-consuming. For example, a team might spend months collecting more training data.
 - This effort is wasted if collecting more data *wasn't* the right solution for the problem.
-

The Solution: Machine Learning Diagnostics 🛠️

Instead of guessing, you should run **diagnostics**.

- A **diagnostic** is a test you can run to gain insight into what is or isn't working with your learning algorithm.
- It provides guidance on which of the options (e.g., "get more data") is most likely to improve your performance.
- **Key Value:** Running a diagnostic (which may take some time to implement) can **save you weeks or even months** of wasted effort by helping you make better decisions.

Before we can use diagnostics, we must first have a clear way to evaluate our learning algorithm's performance.